INVESTIGATION OF ASTROPHYSICAL S-FACTOR

for (α,n) Reactions Using Python

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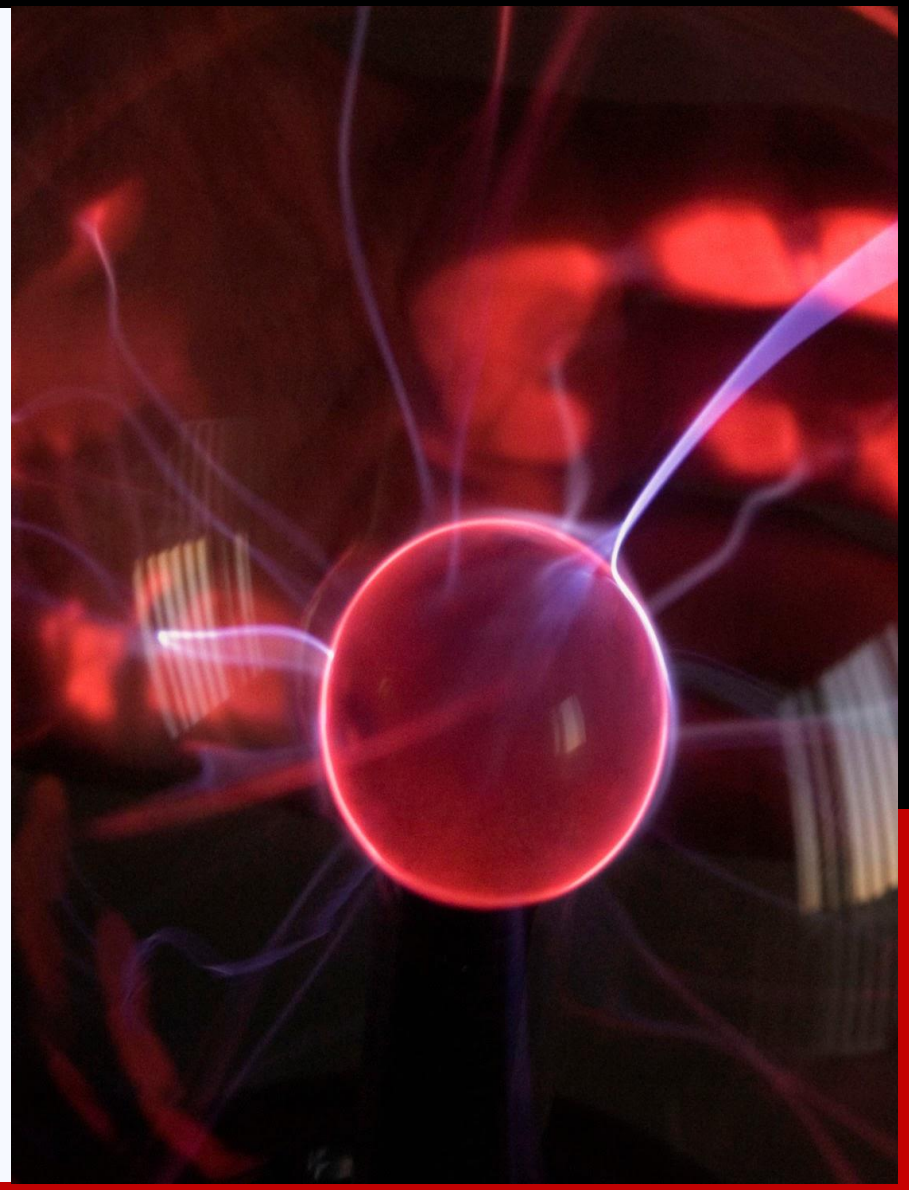


OBJECTIVE AND MOTIVATION

- **Thermonuclear reactions** are the fundamental energy source of stars and drive the formation of heavier elements through **stellar nucleosynthesis**.

- Among them, **(α,n) reactions** play a key role as **neutron sources** in the *slow neutron capture (s-process)* in Asymptotic Giant Branch (AGB) stars.

- **Experimental challenges:** Measuring cross-sections at stellar energies is difficult due to extremely low reaction probabilities and the **Coulomb barrier**.

- **Computational motivation:** Reliable prediction of S-factors and reaction rates using **Python-based numerical methods** helps bridge the gap where experimental data are limited.

Research Objectives

- Compute and compare **S-factors** and **reaction rates** for selected (α,n) reactions using EXFOR data.
- Derive **S-factors** by two methods:
  1. From the relation $S(E) = E\sigma(E)e^{2\pi\eta}$
  2. From **ln(S)** polynomial equations (using anti-log).
- Calculate **reaction rates** through direct integration and from **anti-log** values.
- Analyze differences between both approaches for consistency and accuracy.

Theoretical Foundations

Coulomb Barrier:

- Charged nuclei repel each other, preventing fusion at low energies.
- Probability of tunneling through this barrier is governed by the **Gamow factor**: $P \propto e^{-2\pi\eta}$

where,

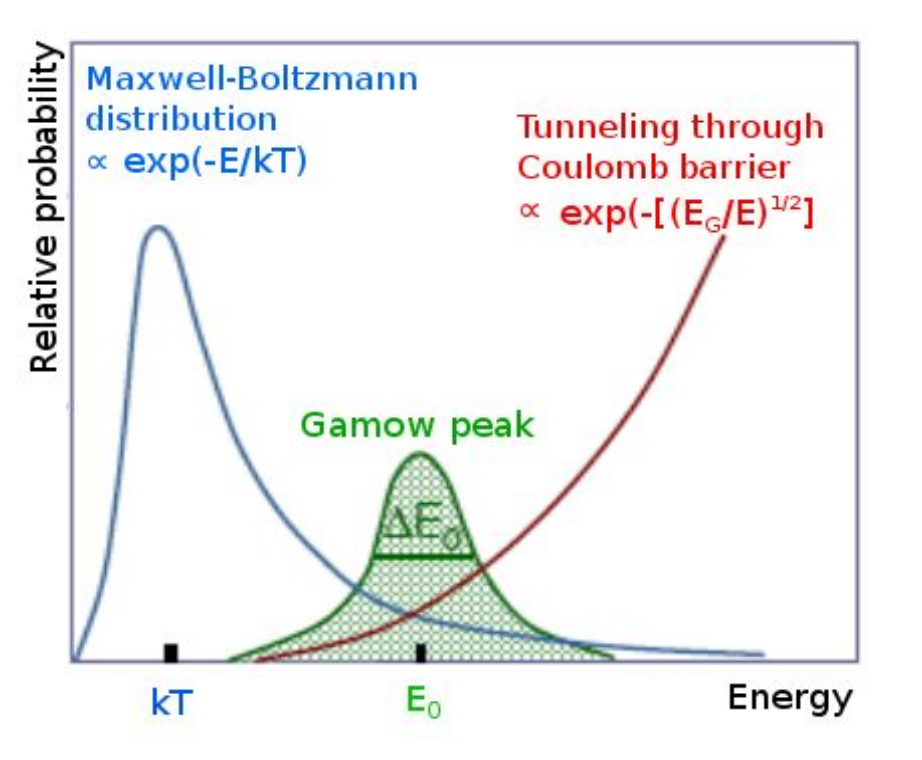
$\eta = \frac{Z_1 Z_2 e^2}{\hbar v}$ — the Sommerfeld parameter.

Astrophysical S-Factor:

- Removes the rapidly varying tunneling effect from the cross-section:

$$S(E) = E\sigma(E)e^{2\pi\eta}$$

- Providing a smoother function useful for extrapolation to stellar energies.



Gamow Window:

- Defines the energy range where reactions in stars are most probable, determined by the overlap of Maxwell-Boltzmann distribution and tunneling probability.

Theoretical Foundations

- **Reaction Rate Framework**

- Reaction rate connects nuclear cross-sections with stellar temperatures through the probability of nuclei overcoming the Coulomb barrier.
- It is given by,

$$N_A\langle\sigma v\rangle = 3.7313 \times 10^7\, \mu^{-1/2} T_9^{-3/2} \int_0^\infty E\sigma(E) e^{-11.605E/T_9}\, dE$$

,where N_A is Avogadro's number and T_9 is the temperature in GK.,σ(E) is cross section(m-barns) with respect to energy(MeV).

- **Purpose**

- Links Eσ(E) with stellar reaction probabilities.
- Comparison of integrated and anti-logarithmic results checks model accuracy.
- Validates the S-factor approach for astrophysical (α,n) reactions.

Data Extraction and Error Analysis

Data Source

- Experimental (α,n) data from **EXFOR** database.
- Includes measured **cross-sections (σ(E))**, **energies**, and **errors** for some of the target nuclei.

$$^{45}\mathrm{Sc}(\alpha,n)^{48}\mathrm{V},$$
$$^{48}\mathrm{Ti}(\alpha,n)^{51}\mathrm{Cr},$$
$$^{51}\mathrm{V}(\alpha,n)^{54}\mathrm{Mn},$$
$$^{50}\mathrm{Cr}(\alpha,n)^{53}\mathrm{Fe},$$
$$^{55}\mathrm{Mn}(\alpha,n)^{58}\mathrm{Co}.$$

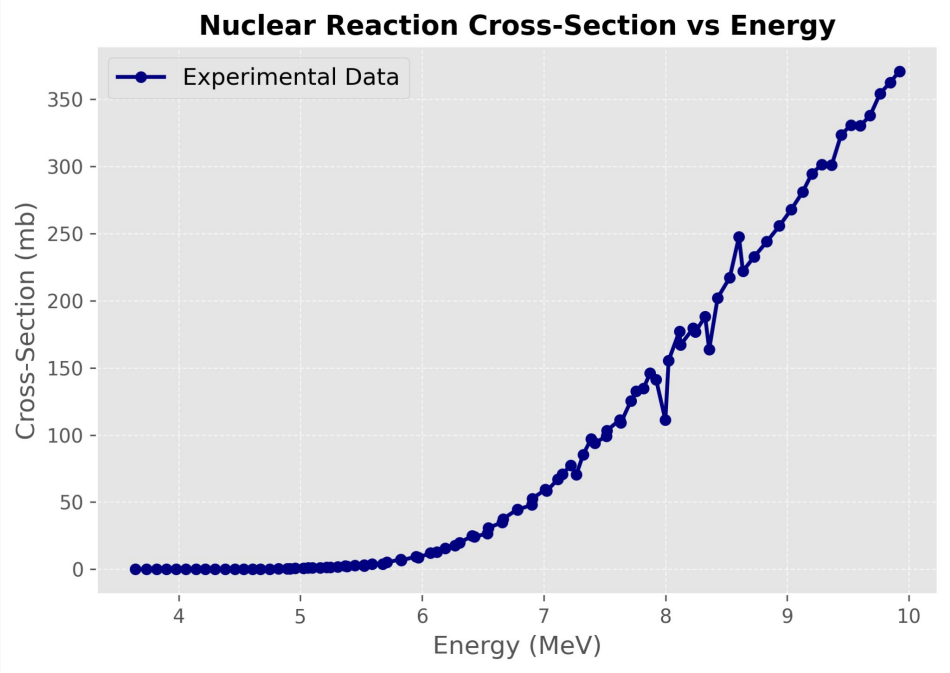
Error Handling

- EXFOR provides statistical and systematic errors in σ(E).
- These are propagated into the S-factor as:

$$\frac{\Delta S(E)}{S(E)} = \frac{\Delta\sigma(E)}{\sigma(E)}$$

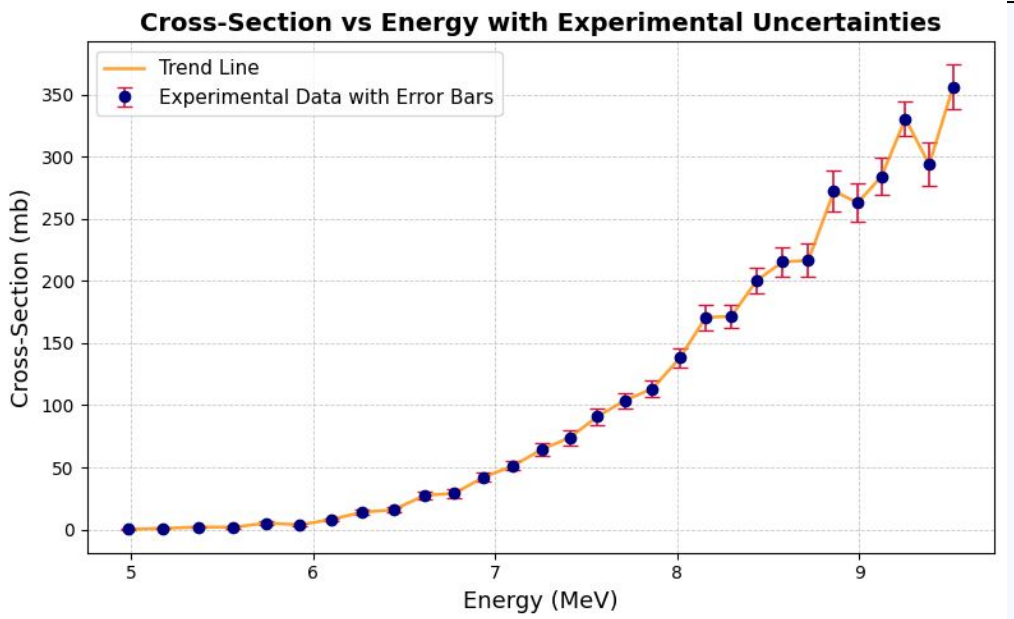
- Ensures consistency between experimental uncertainty and derived S-factor values.

Cross-Section (σ) Dependence on Energy

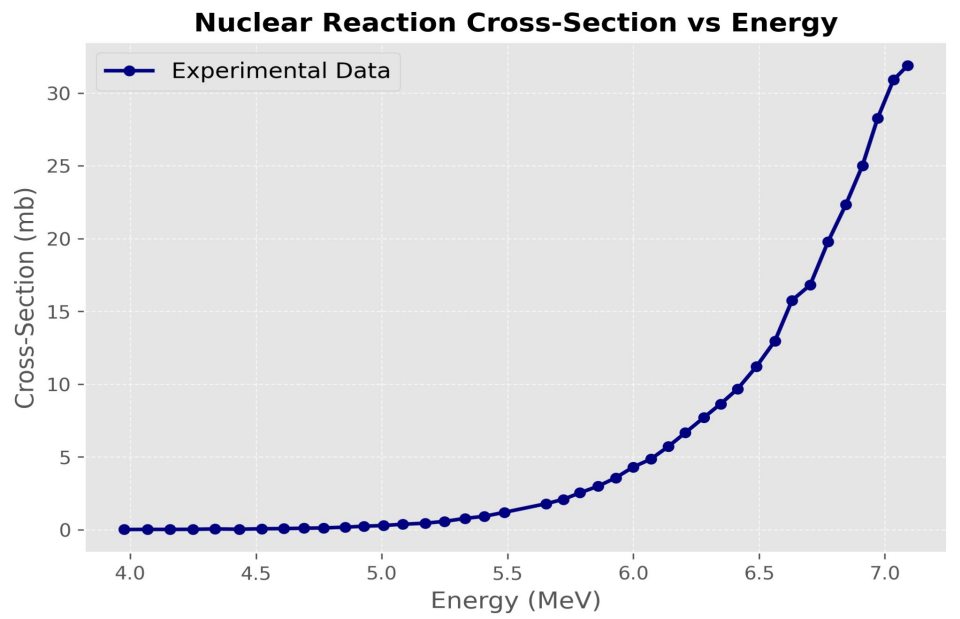


$^{45}\mathrm{Sc}(\alpha, n)^{48}\mathrm{V}$

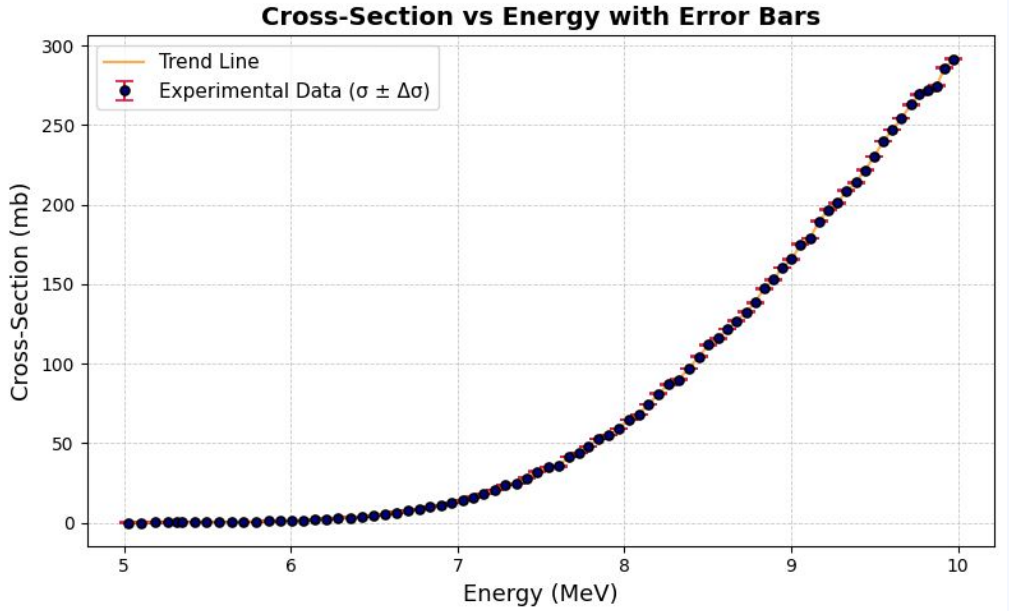
- The experimental cross-section σ(E) quantifies the probability of the (α,n) reaction.
- Data extracted from EXFOR for all five targets: 45Sc, 48Ti, 51V, 50Cr, 55Mn.
- σ(E) shows a gradual increase with energy due to improved tunneling through the Coulomb barrier.
- At higher energies, small resonance-like structures appear.
- Measurement uncertainties are indicated as error bars in each plot.



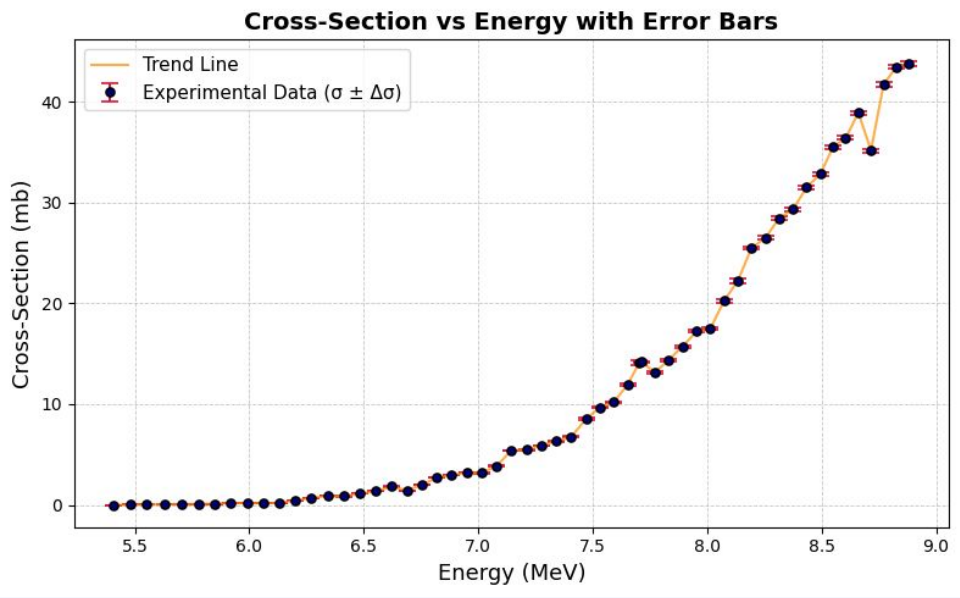
Reaction: $^{48}\mathrm{Ti}(\alpha,n)^{51}\mathrm{Cr}$



Reaction: $^{51}\mathrm{V}(\alpha,n)^{54}\mathrm{Mn}$



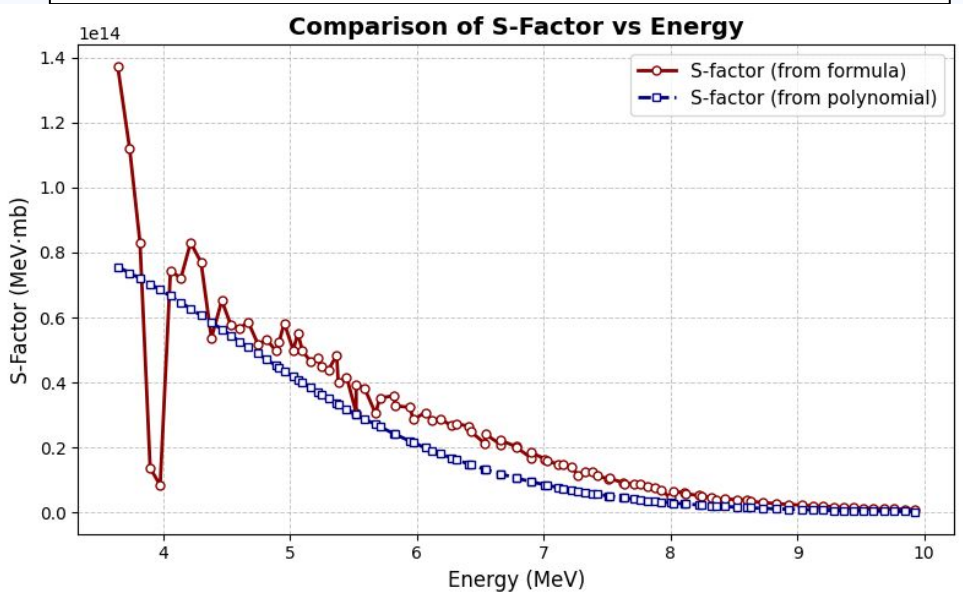
Reaction: $^{50}\mathrm{Cr}(\alpha,n)^{53}\mathrm{Fe}$



Reaction: $^{55}\mathrm{Mn}(\alpha,n)^{58}\mathrm{Co}$

S-Factor vs Energy

Reaction: $^{45}\mathrm{Sc}(\alpha, n)^{48}\mathrm{V}$



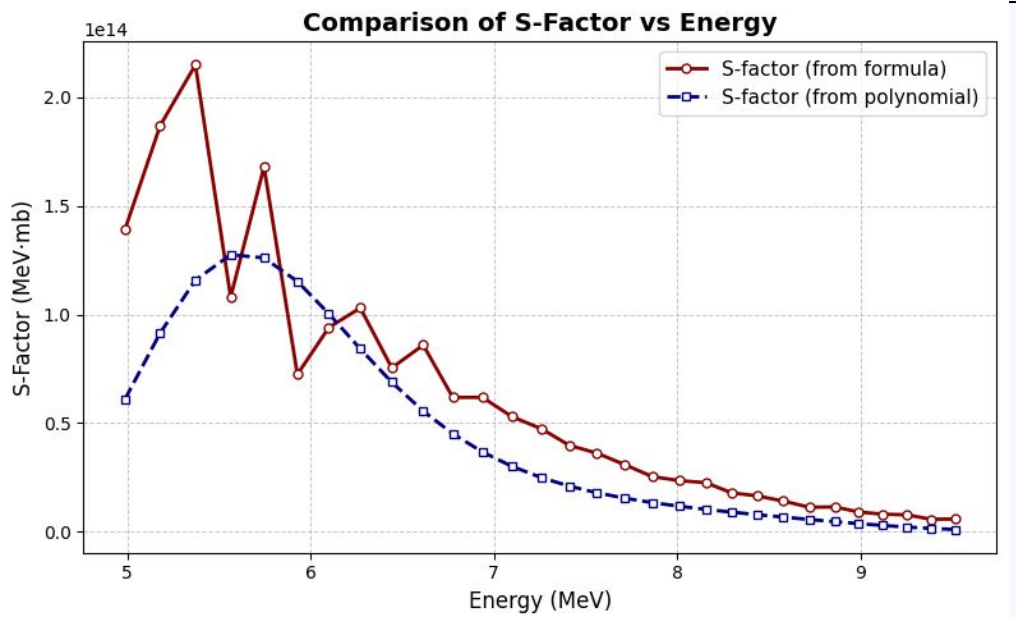
- The astrophysical S-factor smooths the steep energy dependence of σ(E) and isolates nuclear effects.
- Two approaches were used:

1. **Direct computation:** $S(E) = E\sigma(E)e^{2\pi\eta}$
2. **Polynomial fit:** A logarithmic polynomial relation of the form

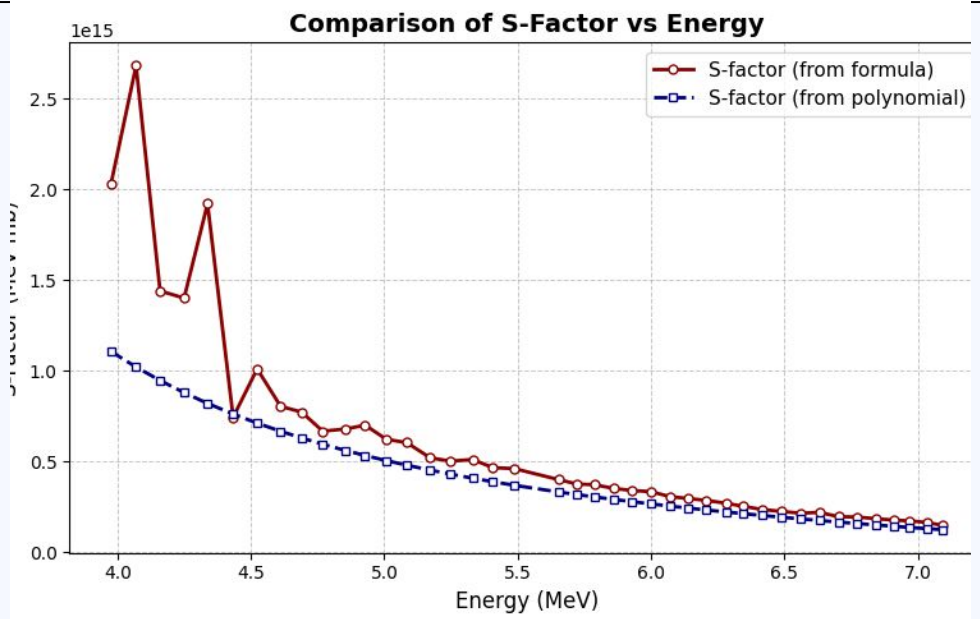
$$\ln[S(E)] = a_0 + a_1 E + a_2 E^2 + a_3 E^3 + \dots$$

followed by **anti-log transformation** to obtain S(E).

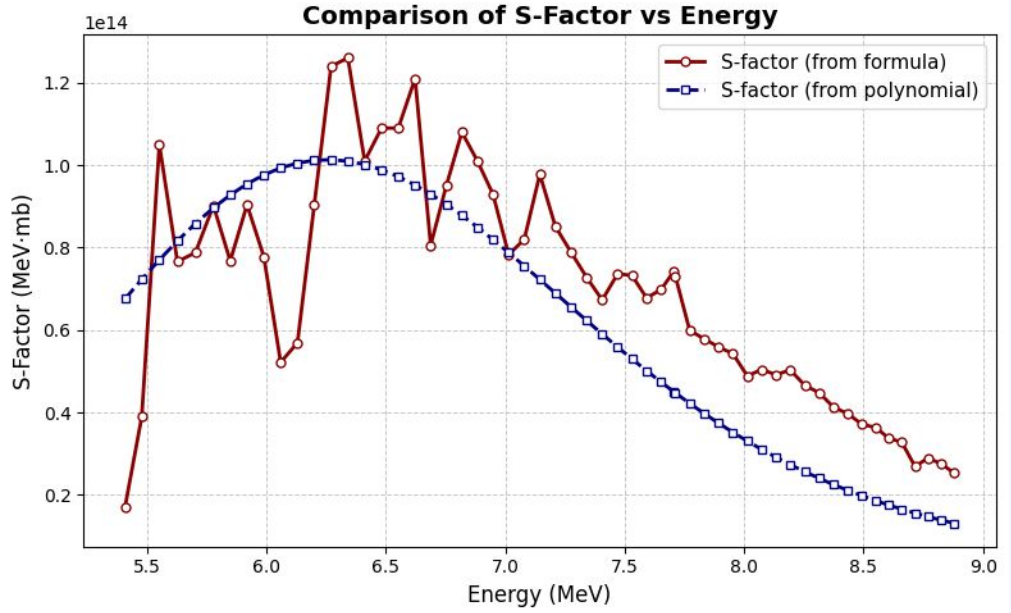
- The polynomial approach ensures a smooth extrapolation where data are sparse.
- Both computed and anti-log-derived S-factors are compared in the plots (5 reactions).



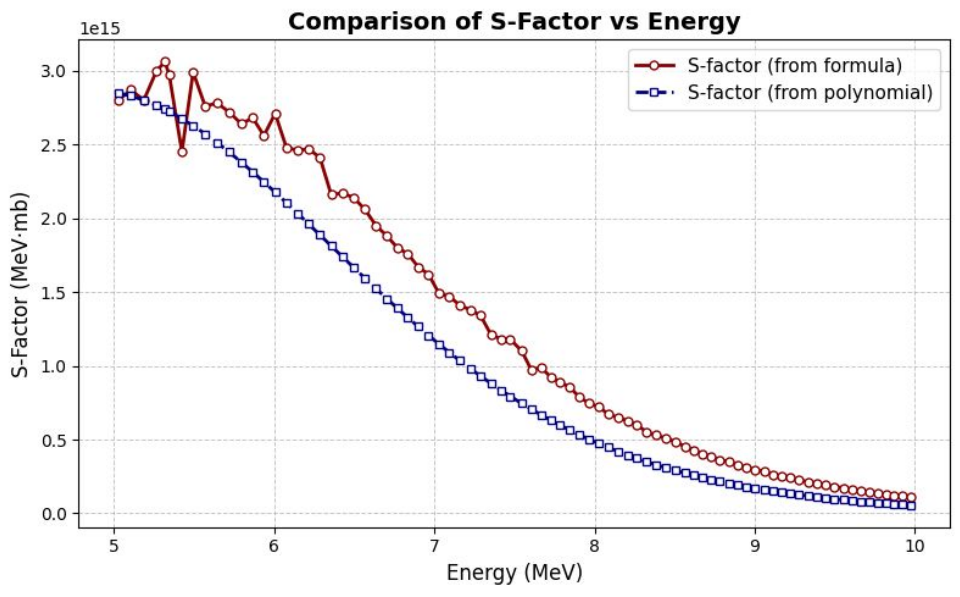
Reaction: $^{48}\mathrm{Ti}(\alpha,n)^{51}\mathrm{Cr}$



Reaction: $^{51}\mathrm{V}(\alpha,n)^{54}\mathrm{Mn}$

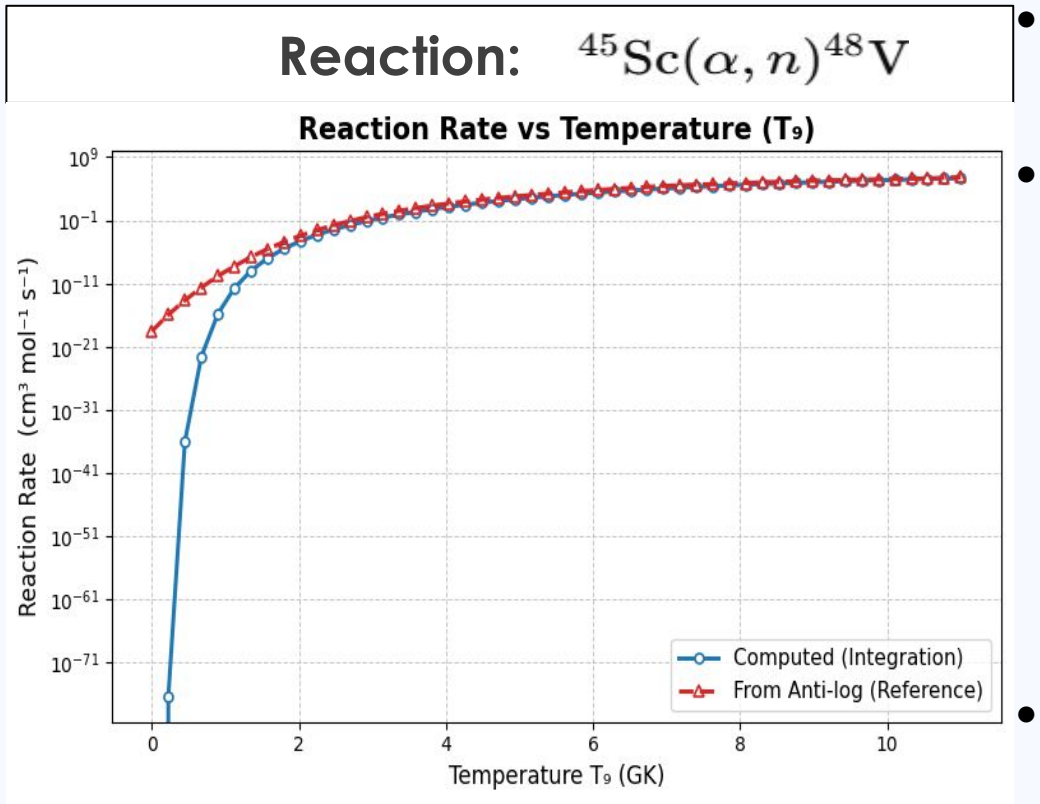


Reaction: $^{50}\mathrm{Cr}(\alpha,n)^{53}\mathrm{Fe}$



Reaction: $^{55}\mathrm{Mn}(\alpha,n)^{58}\mathrm{Co}$

Temperature-Dependent Reaction Rates



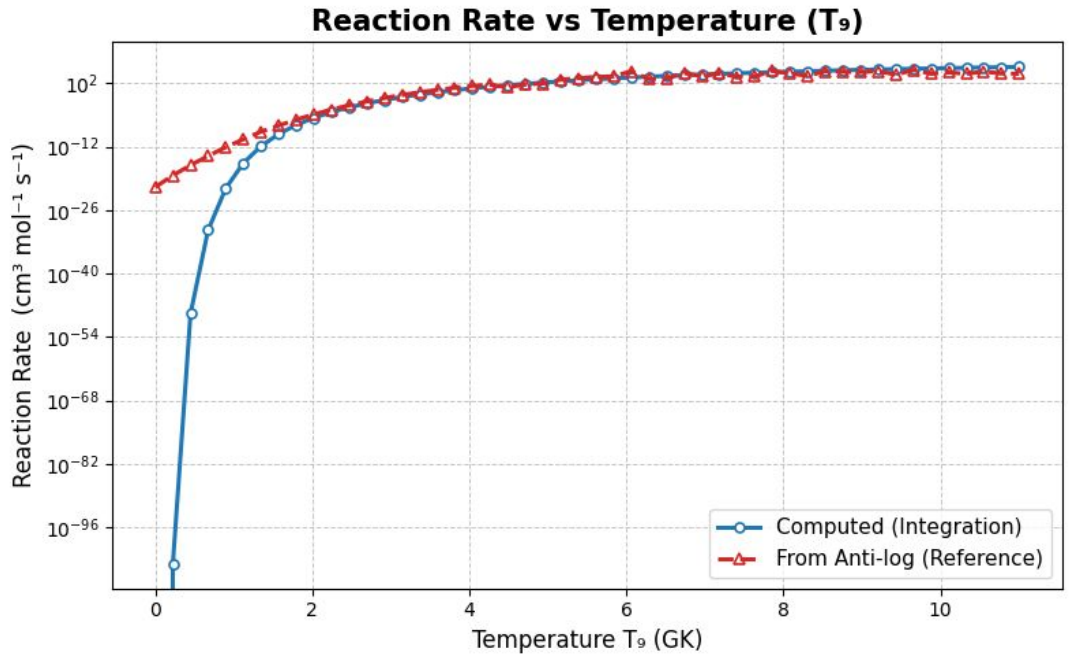
- The thermonuclear reaction rate $N_A\langle\sigma v\rangle$ links nuclear data with stellar temperatures (T_9).
- Two sets of results are shown:
  1. **Integrated values** — obtained directly from the numerical evaluation of ,

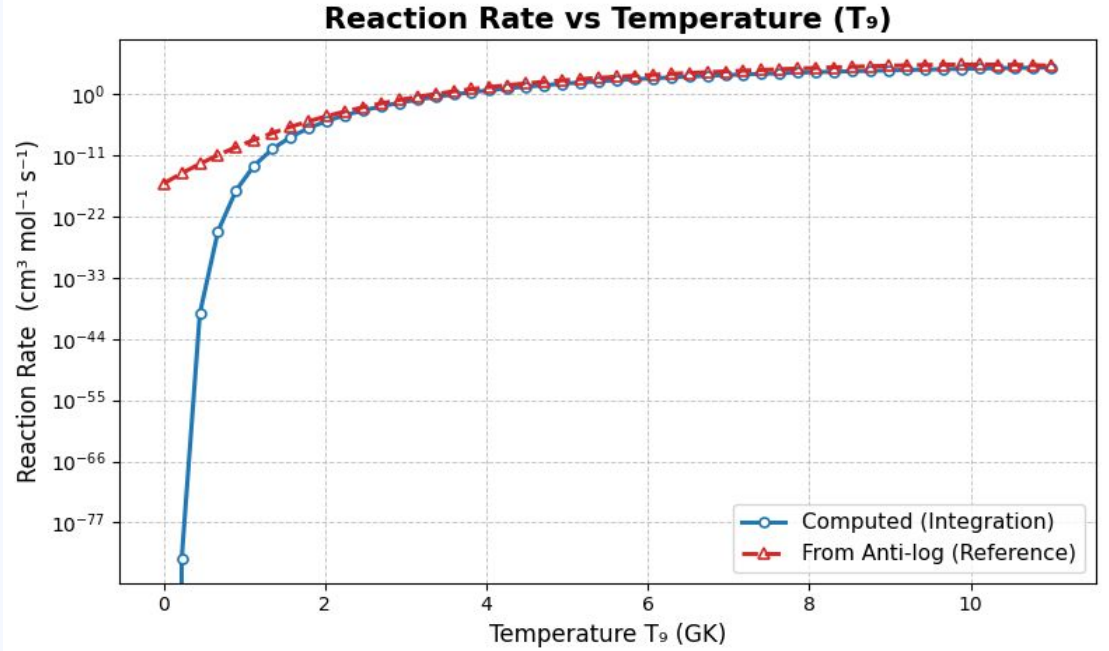
     $$E\sigma(E)e^{-11.605E/T_9}$$

  2. **Anti-log values** — derived from the polynomial expression

     $$\ln[N_A\langle\sigma v\rangle] = b_0 + b_1T_9 + b_2T_9^2 + b_3T_9^3 + \dots$$

     and converted back via anti-logarithm.
- The anti-log results align well with integrated data, confirming consistency across all five reactions.
- Distinct trends appear within the **Gamow window (0.1–1 GK)** where rates rise sharply.


Reaction Rate vs Temperature (T₉)
Reaction Rate (cm³ mol⁻¹ s⁻¹)
Temperature T₉ (GK)
Computed (Integration)
From Anti-log (Reference)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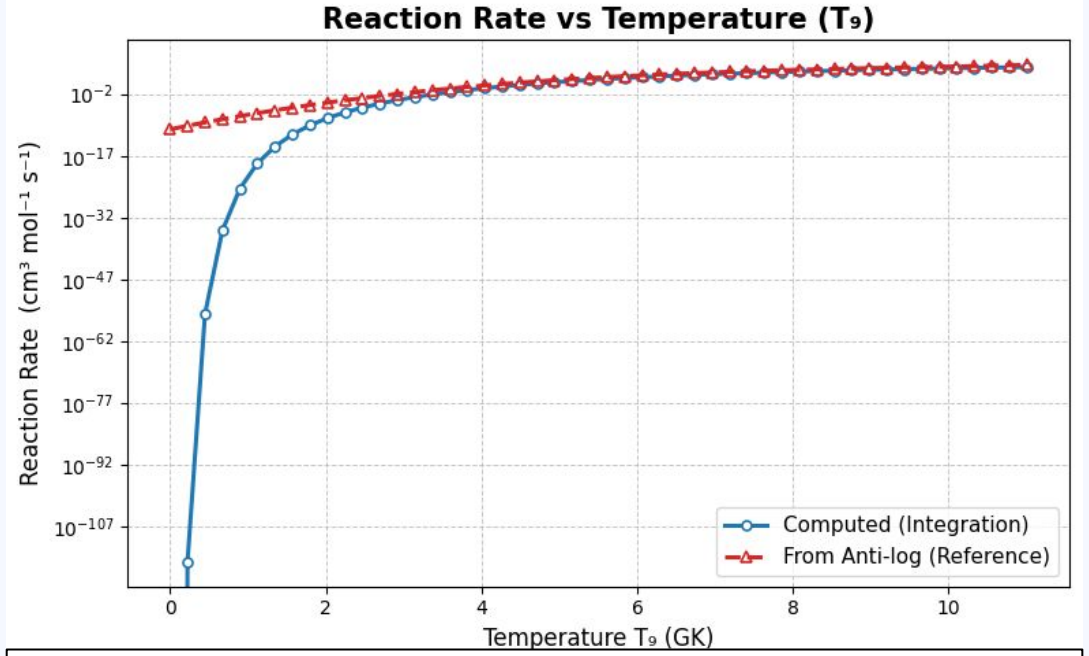
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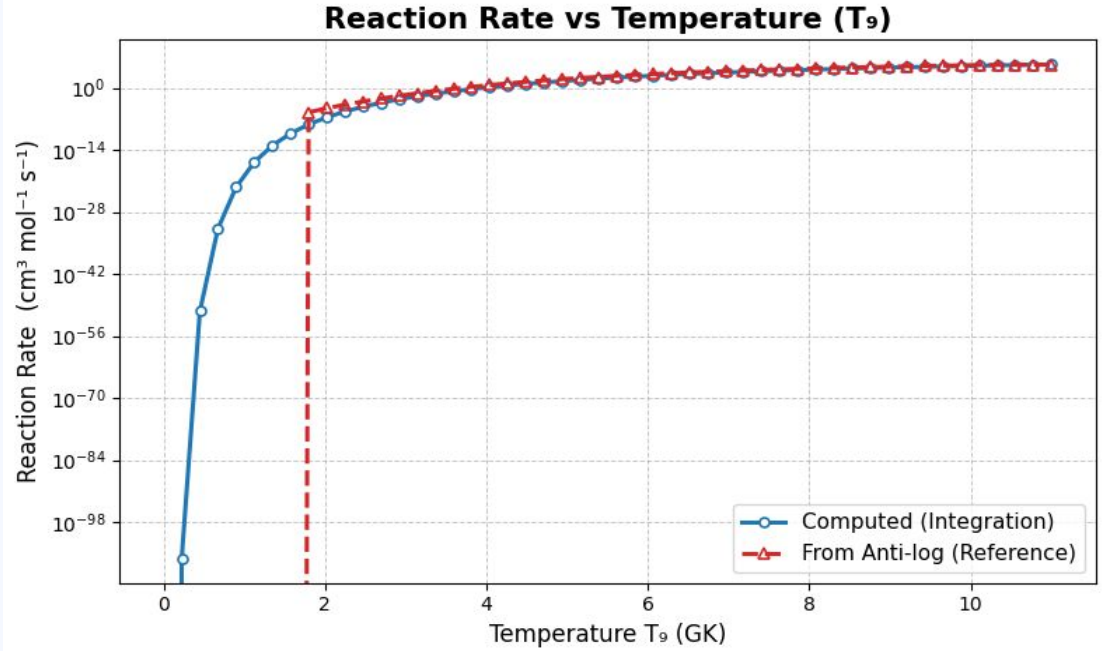
Reaction Rate vs Temperature (T₉)
Reaction Rate (cm³ mol⁻¹ s⁻¹)
Temperature T₉ (GK)
Computed (Integration)
From Anti-log (Reference)

Key Findings and Future Scope

Key Findings:

- Computed cross-sections, S-factors, and reaction rates for five (α,n) reactions.
- S-factor from both formula and polynomial shows smooth, consistent behavior.
- Reaction rates rise exponentially with temperature, peaking in the Gamow window.
- Integrated and anti-log results closely agree, validating computational accuracy.

Future Scope:

- Extend analysis to other (α,n) and (p,γ) reactions.
- Include uncertainty and error propagation in future models.
- Compare with theoretical predictions and refine integration methods.

Conclusion

- The computational approach effectively models astrophysical (α,n) reactions.
- S-factors derived from both experimental data and polynomial fits show consistent trends.
- Reaction rate analysis confirms temperature dependence matching theoretical expectations.
- The study strengthens the link between nuclear data and stellar nucleosynthesis modeling.
- Demonstrates the reliability of Python-based computation for nuclear astrophysics research

Acknowledgement

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thank you